

The Volume-Independent DET16 Mirror-Gap Theorem: Exact Product Gap, Spin(10) Symmetry, and the Honest Stability Threshold

Proof memorandum

31 August 2026

Abstract

On every finite set of disjoint sixteen-mode clusters we prove exactly that the DET16 Hamiltonian

$$H_0 = \sum_x (\mathbf{1} - P_{\phi,x})$$

has the unique product ground state $\bigotimes_x (|0\rangle_x + e^{i\phi}|F\rangle_x)/\sqrt{2}$, spectrum $\{0, 1, \dots, |\Lambda|\}$, and gap one independently of volume. The proof includes the graded-CAR argument: the determinant monomial has even degree sixteen, so local projectors on disjoint clusters commute without an ordering sign. We also prove local Spin(10) Gauss invariance, fermion parity, translation symmetry, absence of spontaneous symmetry breaking, and absence of torus degeneracy.

For a bounded finite-range even perturbation, the Michalakis–Zwolak stability theorem applies because frustration freeness, Local-TQO, and the local gap are exact, with $\Delta_0(\ell) = 0$ and $\gamma(r) = 1$. It gives a volume-independent threshold $t_* > 0$ and perturbed gap at least $1/2$. For finite $\mathbb{Z}_2/\mathbb{Z}_4$ dynamical links on an open chain, the Fröhlich–Pizzo Lie–Schwinger theorem applies after an exact ambient-cell and endpoint repair. It gives a unique physical ground state and gap at least $\Delta/2$, uniformly in chain length, for $|t| < \Delta\tau_{\text{FP}}/m$, where $\Delta = \min(1, h_E)$ and $\tau_{\text{FP}} > 0$ is only partially explicit in the cited proof. It does *not* give the proposed numerical threshold $1/(2 \cdot 16z)$ for ordinary hopping. The numeric twins pass all checks and find no finite-volume closing in their declared scans. The final box now isolates compact-group rotor links, wall mixing, and the three-spatial-dimensional placement; none is included in the finite-group chain theorem.

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1 Concrete model and frozen finite input

1.1 One DET16 cluster

Let $K \cong \mathbb{C}^{16}$ carry the positive-chirality spin representation ρ_{16} of $\text{Spin}(10)$ and let $\mathcal{F}(K) = \bigoplus_{n=0}^{16} \wedge^n K$ be its fermionic Fock space. For an ordered orthonormal basis $(e_1, \dots, e_{16})$, write

$$\Omega^\dagger = a^\dagger(e_1) \cdots a^\dagger(e_{16}), \quad |F\rangle = \Omega^\dagger |0\rangle,$$

and $\Omega = (\Omega^\dagger)^*$. Put

$$\begin{aligned} A_\phi &= P_0 + P_{16} + e^{-i\phi} \Omega + e^{i\phi} \Omega^\dagger, \\ |\Omega_\phi\rangle &= \frac{|0\rangle + e^{i\phi} |F\rangle}{\sqrt{2}}, \\ P_\phi &= \frac{1}{2} A_\phi = |\Omega_\phi\rangle \langle \Omega_\phi|, \quad q_\phi = \mathbf{1} - P_\phi. \end{aligned} \tag{1.1}$$

The exact finite computation in `verification/v1002_spin10_mirror_projector.py` verifies

$$\Omega^\dagger \Omega = P_{16}, \quad \Omega \Omega^\dagger = P_0, \quad P_\phi^2 = P_\phi = P_\phi^*, \quad \text{rank } P_\phi = 1, \tag{1.2}$$

and all forty-five $\mathfrak{so}(10)$ commutators vanish. These are inputs to this memorandum; the algebraic consequences needed below are also proved directly.

1.2 Many disjoint clusters

Let Λ be any nonempty finite set of clusters. Fix an ordering only to identify the global Fock space with an ordinary tensor product,

$$\mathcal{F}\left(\bigoplus_{x \in \Lambda} K_x\right) \cong \widehat{\bigotimes_{x \in \Lambda} \mathcal{F}(K_x)}, \tag{1.3}$$

where the hat recalls that the intrinsic product is graded. Operators have fermion parity $|A| \in \{0, 1\}$. If A_x and B_y are homogeneous and supported on disjoint clusters, the CAR give

$$A_x B_y = (-1)^{|A_x||B_y|} B_y A_x. \tag{1.4}$$

The terms $P_{0,x}$ and $P_{16,x}$ preserve particle number, while Ω_x and $\Omega_x^\dagger$ contain sixteen odd generators. Every summand of $P_{\phi,x}$ is therefore even. Hence

$$[P_{\phi,x}, P_{\phi,y}] = [q_{\phi,x}, q_{\phi,y}] = 0 \quad (x \neq y) \tag{1.5}$$

with no ordering-dependent sign. Define

$$H_{0,\Lambda} = \sum_{x \in \Lambda} q_{\phi,x}. \tag{1.6}$$

Unless stated otherwise, the same phase ϕ is used at every site.

2 Exact spectrum and the volume-independent unit gap

Lemma 2.1 (Order independence of even local products). *For any subset $S \subseteq \Lambda$, the product $\prod_{x \in S} P_{\phi,x}$ is independent of the order chosen for its factors. Under any ordered tensor-product identification (1.3), it is the ordinary tensor product of the local rank-one projectors on S and the identity on S^c .*

Proof. Each $P_{\phi,x}$ is even. Equation (1.4) therefore has sign $(-1)^{0 \cdot 0} = 1$ for every transposition of two factors. Any two orderings differ by transpositions, proving the first claim. An even operator crosses all preceding graded tensor factors without a Jordan–Wigner parity string. Consequently its representative is the ordinary local matrix on its own factor and identities elsewhere. Multiplying these representatives proves the second claim. $\square$

Theorem 2.2 (T1: exact DET16 product spectrum). *Let $N = |\Lambda|$ and $d = \dim \mathcal{F}(\mathbb{C}^{16}) = 2^{16}$. The commuting Hamiltonian (1.6) has*

$$\text{spec}(H_{0,\Lambda}) = \{0, 1, \dots, N\}. \quad (2.1)$$

The multiplicity of k is

$$\binom{N}{k} (d-1)^k. \quad (2.2)$$

Its unique normalized ground state is

$$|\Omega_{\Lambda,\phi}\rangle = \widehat{\bigotimes_{x \in \Lambda}} |\Omega_{\phi}\rangle_x, \quad (2.3)$$

and

$$\text{gap}(H_{0,\Lambda}) = 1 \quad (2.4)$$

for every finite volume and every boundary condition.

Proof. On one cluster, $\mathcal{F}(K_x) = \text{Ran } P_{\phi,x} \oplus \text{Ran } q_{\phi,x}$ with dimensions 1 and $d-1$. By Lemma 2.1, the global Hilbert space is the orthogonal direct sum

$$\bigoplus_{S \subseteq \Lambda} \left(\bigotimes_{x \in S} \text{Ran } q_{\phi,x} \otimes \bigotimes_{y \notin S} \text{Ran } P_{\phi,y} \right). \quad (2.5)$$

On the summand labelled by S , exactly the projectors $q_{\phi,x}$ with $x \in S$ act as one and all others act as zero. Thus $H_{0,\Lambda}$ acts as the scalar $|S|$. This proves (2.1). For $|S| = k$ there are $\binom{N}{k}$ choices of S , and every chosen excited factor has dimension $d-1$, proving (2.2).

At energy zero, $S = \emptyset$ is the only summand and every local factor lies in the one-dimensional range of $P_{\phi,x}$. Its normalized vector is exactly (2.3); evenness makes the product independent of the temporary cluster ordering. The next spectral value is one, proving (2.4). No edge or torus identification enters (1.6), so the conclusion is independent of boundary condition. $\square$

3 Gauge, parity, translations, and degeneracy

Lemma 3.1 (The chiral determinant character is trivial). *For every $g \in \text{Spin}(10)$,*

$$\det \rho_{16}(g) = 1. \quad (3.1)$$

Proof. The map $\chi = \det \circ \rho_{16} : \text{Spin}(10) \rightarrow U(1)$ is a continuous group character. Its differential at the identity is

$$d\chi(X) = \text{Tr}(d\rho_{16}(X)).$$

The trace vanishes on commutators. Since $\mathfrak{spin}(10)$ is semisimple, $[\mathfrak{spin}(10), \mathfrak{spin}(10)] = \mathfrak{spin}(10)$, so this linear functional vanishes identically. Thus χ is locally constant. The group $\text{Spin}(10)$ is connected, and $\chi(e) = 1$, hence $\chi(g) = 1$ for every g . $\square$

Theorem 3.2 (T2: exact symmetries and no hidden degeneracy). *The Hamiltonian $H_{0,\Lambda}$ has the following properties.*

- (a) *It is invariant under independent local $\text{Spin}(10)$ transformations. Equivalently, for every matter Gauss generator $G_x^A = d\Gamma_x(T^A)$,*

$$[H_{0,\Lambda}, G_x^A] = 0 \quad (x \in \Lambda, A = 1, \dots, 45). \quad (3.2)$$

- (b) *It commutes with total fermion parity. If ϕ is constant, it is invariant under every lattice translation preserving Λ and its boundary condition.*
- (c) *The finite-volume ground state is a symmetry singlet. The infinite-volume product state is the unique frustration-free ground state, so none of these symmetries is spontaneously broken.*
- (d) *On a torus the ground-state degeneracy is exactly one. In particular, the model has no topological ground-state degeneracy.*

Proof. Second quantization fixes the vacuum. On the filled vector it acts by the top exterior power:

$$\Gamma(\rho_{16}(g))|F\rangle = (\det \rho_{16}(g))|F\rangle = |F\rangle$$

by Lemma 3.1. Hence $|\Omega_\phi\rangle$, P_ϕ , and q_ϕ are invariant. This is true independently at every cluster, so differentiation gives (3.2). The forty-five zero commutators in the frozen finite computation are the matrix twin of this argument.

Both components of $|\Omega_\phi\rangle$ have even parity because their particle numbers are 0 and 16. Thus every $P_{\phi,x}$ and $q_{\phi,x}$ is even and H_0 commutes with total parity. A translation merely permutes identical terms when ϕ is constant, which proves translation invariance.

The product ground vector is fixed by all the stated symmetries. For the infinite system, let ω be any frustration-free state. Then $\omega(q_{\phi,x}) = 0$ for every x . Positivity and $q_{\phi,x} \geq 0$ imply that the one-site reduced state is supported on $\ker q_{\phi,x} = \mathbb{C}|\Omega_\phi\rangle$. A state with a pure one-site marginal factorizes across that site; iterating over every finite region gives

$$\omega|_A = \bigotimes_{x \in A} |\Omega_\phi\rangle\langle\Omega_\phi|$$

for every finite A . Local restrictions determine a quasi-local state, so ω is the unique product state. There are therefore no distinct infinite-volume ground states related by a symmetry and no spontaneous symmetry breaking.

Finally, Theorem 2.2 applies unchanged on a torus and gives a one-dimensional ground space. More concretely, all local constraints $q_{\phi,x} = 0$ force the same rank-one factor independently at every site; no noncontractible cycle supplies an additional label. This proves the torus claim directly from the commuting-projector structure. $\square$

4 The classical stability theorem, stated in full

4.1 Local-TQO and local gap

For a finite region $B \subseteq \Lambda$, define

$$H_B = \sum_{x \in B} q_{\phi, x}, \quad P_B = \prod_{x \in B} P_{\phi, x}. \quad (4.1)$$

As an operator on the full system, P_B is the identity outside B . If $A \subset B$ and O_A is supported on A , then

$$P_B O_A P_B = \langle \Omega_{A, \phi}, O_A \Omega_{A, \phi} \rangle P_B. \quad (4.2)$$

Thus local ground states are exactly indistinguishable in the interior: the Local-TQO error function is identically zero. Theorem 2.2 applied to B also gives

$$\text{gap}(H_B) = 1 \quad (4.3)$$

for every nonempty B .

Definition 4.1 (Finite-range perturbation strength). *An even finite-range perturbation is a sum*

$$V_\Lambda = \sum_{X \subseteq \Lambda} V_X, \quad V_X = V_X^*, \quad |V_X| = 0, \quad (4.4)$$

for which $V_X = 0$ when $\text{diam } X > R$ and

$$\sup_{x \in \Lambda} \sum_{X \ni x} \|V_X\| \leq J_V \quad (4.5)$$

uniformly in Λ . Gauge covariance means that the complete local term, including any prescribed parallel transporter, is $\text{Spin}(10)$ -invariant. Evenness makes disjoint interaction terms commute in the graded sense required by the local stability argument.

Theorem 4.2 (Michalakis–Zwolak stability theorem, full form used here). *Let H_0 be a finite-range frustration-free Hamiltonian on finite subsets of $\mathbb{Z}^D$, with a volume-independent spectral gap $\gamma > 0$. For every ball $b_u(r)$ let $P_{b_u(r)}$ be the ground projection of the restricted Hamiltonian. Assume:*

- (i) Local-TQO. *For every ball $A = b_u(r)$, every bounded operator O_A supported in A , and $A(\ell) = b_u(r + \ell)$,*

$$\left\| P_{A(\ell)} O_A P_{A(\ell)} - \frac{\text{Tr}(P_{A(\ell)} O_A)}{\text{Tr } P_{A(\ell)}} P_{A(\ell)} \right\| \leq \|O_A\| \Delta_0(\ell), \quad (4.6)$$

where $\Delta_0(\ell)$ decays sufficiently rapidly and the admissible topological-order length grows as a positive power of the system diameter.

- (ii) Local gap. *Every restricted Hamiltonian on $b_u(r)$ has no spectrum in $(0, \gamma(r))$, where $\gamma(r) > 0$ decays at most polynomially in r .*
- (iii) Quasi-local perturbation. *The perturbation is a (J, f_0) interaction: it is a sum of bounded terms centred at lattice sites whose range- r norms are bounded by $J f_0(r)$, with f_0 decaying rapidly enough for a Lieb–Robinson bound.*

Then there exist constants $J_0 > 0$ and $L_0 \geq 2$, independent of the finite volume, such that whenever $J \leq J_0$ and the system diameter $L \geq L_0$,

$$\text{gap}(H_0 + V) \geq \frac{\gamma}{2}. \quad (4.7)$$

In the proof, after quasi-local spectral-flow block diagonalization, the transformed interaction W obeys

$$|\langle \psi, W\psi \rangle| \leq cJ \langle \psi, H_0\psi \rangle, \quad (4.8)$$

where one may take

$$c = C_D \sum_{k=1}^M \frac{r_k^D}{\gamma(r_k)} \sum_{r_{k-1} < r \leq r_k} w(r), \quad 0 = r_0 < r_1 \leq \dots \leq r_M = L, \quad (4.9)$$

C_D is the volume of the unit ball in $\mathbb{Z}^D$, and w is the rapid-decay function of the transformed interaction. The final bootstrap may choose

$$J_0 = \frac{1}{3c}. \quad (4.10)$$

The constants w, c, J_0 depend on the interaction-decay and spectral-flow estimates, but not on volume.

This is Theorem 1 and Proposition 2 of S. Michalakis and J. P. Zwolak, *Stability of Frustration-Free Hamiltonians*, Communications in Mathematical Physics **322** (2013), 277–302. It extends the commuting-projector theorem of Bravyi, Hastings, and Michalakis. We have included the hypotheses, conclusion, and the relative-bound constant used in their proof so that no numerical threshold is hidden.

5 Volume-independent perturbative stability

Theorem 5.1 (T3: proved volume-independent stability). *Let Λ range over finite subsets or tori of a fixed D -dimensional bounded-degree lattice. Let $H_{0,\Lambda}$ be the DET16 Hamiltonian (1.6), and let V_Λ be an even finite-range interaction satisfying Definition 4.1 uniformly in volume. Then there exists $t_* > 0$, depending only on D , the interaction range, and the local interaction-norm bound, but not on Λ , such that*

$$\text{gap}(H_{0,\Lambda} + tV_\Lambda) \geq \frac{1}{2} \quad (5.1)$$

for every sufficiently large volume and $|t| \leq t_*$. After decreasing t_* once to cover the finitely many smaller volumes, the same statement holds for every finite Λ .

Proof. Every local term $q_{\phi,x}$ is positive and annihilates the product state, so H_0 is frustration free. Theorem 2.2 gives global gap $\gamma = 1$. Equation (4.2) verifies the full Local-TQO condition (4.6) with

$$\Delta_0(\ell) = 0$$

at every scale, not merely asymptotically. Equation (4.3) verifies the Local-Gap condition with $\gamma(r) = 1$. A finite-range interaction has a compactly supported, hence rapidly decaying, function f_0 . The local on-cluster Hilbert space is finite dimensional. Finally, all perturbation terms are even, so spacelike-separated local terms commute rather than anticommute; the locality and Lieb–Robinson estimates used in Theorem 4.2 apply to this even CAR interaction.

Apply Theorem 4.2 to the perturbation tV . Its strength is $J = |t|J_V$, so one may set $t_* = J_0/J_V$. This gives (5.1) for $L \geq L_0$. There are only finitely many lattice shapes below

L_0 up to the fixed local geometry. Each has a simple isolated ground eigenvalue and gap one at $t = 0$; finite-matrix eigenvalues are continuous in t . Taking the minimum of their positive continuity radii and the large-volume value gives one positive threshold for all finite volumes. $\square$

6 Dynamical finite gauge links: Lie–Schwinger gap theorem

Theorem 5.1 applies to prescribed parallel transporters but does not itself add fluctuating link Hilbert spaces. We therefore audit the finite-group dynamical-link extension first against the local Schrieffer–Wolff construction of Bravyi, DiVincenzo, and Loss [3], and then against the gap theorem proved by Fröhlich and Pizzo [4].

Let each oriented open-chain link carry the electric basis of $\mathbb{C}[\mathbb{Z}_N]$, $N = 2$ or 4 , and let U_ℓ shift its electric flux. On the unconstrained ambient tensor product put

$$H_{0,\text{dyn}} = \sum_x (\mathbf{1} - P_{\phi,x}) + h_E \sum_\ell E_\ell^2, \quad V = \sum_{\ell=(x,x+1)} K_\ell, \quad (5.2)$$

where

$$K_\ell = \sum_{\alpha=1}^m \left(c_{x,\alpha}^\dagger U_\ell c_{x+1,\alpha} + c_{x+1,\alpha}^\dagger U_\ell^\dagger c_{x,\alpha} \right). \quad (5.3)$$

Here $m = 16$ for DET16 and $m = 4$ in the numeric analogue. Assigning each outgoing link to its left matter cell makes (5.2) a sum of one-cell terms and (5.3) a two-cell interaction. Its product ground projector is the tensor product of the DET projectors and the zero-electric-flux link projectors, and its local gap is

$$\Delta = \min(1, h_E). \quad (5.4)$$

Moreover $\|K_\ell\| = m$, so the BDL interaction strength obeys

$$J = \max_x \sum_{\ell \ni x} \|K_\ell\| \leq 2m \quad (5.5)$$

uniformly in chain length.

Proposition 6.1 (BDL hypothesis repair and remaining obstruction). *For finite $\mathbb{Z}_2$ or $\mathbb{Z}_4$ links, the Gauss-restricted Hilbert space is not a tensor product and hence is not directly in the setup of Section 4 of Bravyi–DiVincenzo–Loss. The ambient-cell formulation (5.2)–(5.5) repairs this locality hypothesis, and gauge invariance makes every Gauss sector reducing. Nevertheless the cited BDL result does not supply the requested volume-independent spectral-gap theorem or a numerical threshold.*

Proof. The local Gauss constraints correlate a matter cell with its incident links, so the physical subspace does not factor into local Hilbert spaces. On the ambient space, by contrast, the left-link cell assignment gives exactly the tensor product, one-cell unperturbed Hamiltonian, product ground space, and two-cell perturbation assumed in BDL Section 4.1. The determinant changes charge by sixteen and the dressed hopping transports one unit of link flux with the matter charge, so all terms commute with every Gauss generator. Restriction to the physical sector therefore preserves the spectrum originating from its invariant subspace. Adding a Gauss penalty is unnecessary; it would produce overlapping commuting local terms and would no longer have BDL’s one-site unperturbed form.

For a second-order truncation $n = 2$, BDL Theorem 3 normalizes $\|V\|_1 = 1$ and obtains

$$\epsilon_c = \min \left\{ \frac{\Delta\beta}{4n^2}, \frac{\Delta}{32\alpha} \right\}, \quad (5.6)$$

where α and β arise from big- O estimates in their Lemma 4.2 and are not evaluated numerically. Writing $\epsilon = tJ$ and using (5.5) gives only the symbolic conditions

$$\begin{aligned} m = 4 : \quad |t| &< \Delta \min \left\{ \frac{\beta}{128}, \frac{1}{256\alpha} \right\}, \\ m = 16 : \quad |t| &< \Delta \min \left\{ \frac{\beta}{512}, \frac{1}{1024\alpha} \right\}. \end{aligned} \tag{5.7}$$

No concrete number follows without values of α, β . More decisively, BDL Theorem 3 controls the *ground-state energy* with an extensive truncation error $O(|\Lambda||\epsilon|^{n+1})$; it does not control the low-sector spectral gap. The paragraph following their Lemma 4.1 explicitly states that the lemma says nothing about that gap. Consequently (5.7) cannot be promoted to a uniform lower bound on $\text{gap}(H_{0,\text{dyn}} + tV)$. $\square$

Theorem 6.2 (Fröhlich–Pizzo Lie–Schwinger theorem, full form used here). *Let $\mathcal{H}^{(L)} = \bigotimes_{j=1}^L \mathcal{H}_j$ with $\mathcal{H}_j \simeq \mathbb{C}^M$, where $M < \infty$ is independent of L . Suppose the identical one-site operators $H_j \geq 0$ have a one-dimensional kernel $\mathbb{C}\Omega_j$ and satisfy*

$$H_j|_{\{\mathbb{C}\Omega_j\}^\perp} \geq \mathbf{1}. \tag{5.8}$$

Let

$$K_L(\tau) = \sum_{j=1}^L H_j + \tau \sum_{\substack{I_{k;i} \subset \{1, \dots, L\} \\ k \leq \bar{k}}} V_{I_{k;i}}, \tag{5.9}$$

where $\bar{k} < \infty$ is fixed independently of L , every $V_{I_{k;i}}$ is self-adjoint and supported on its indicated interval, and $\|V_{I_{k;i}}\| \leq 1$. Then there is $\tau_{\text{FP}} > 0$, independent of L , such that for every real $|\tau| < \tau_{\text{FP}}$ and every finite L , $K_L(\tau)$ has a unique ground state and

$$\text{gap } K_L(\tau) \geq \frac{1}{2}. \tag{5.10}$$

This is Theorem 3.5 of J. Fröhlich and A. Pizzo, *Lie–Schwinger Block-Diagonalization and Gapped Quantum Chains*, Communications in Mathematical Physics **375** (2020), 2039–2069 [4]. Its conclusion is a spectral-gap estimate, not a ground-energy truncation estimate, and is uniform in chain length.

Proposition 6.3 (Lie–Schwinger hypothesis audit for finite gauge links). *For $N_g \in \{2, 4\}$, $h_E > 0$, $m \in \{4, 16\}$, and constant ϕ , the open-chain Hamiltonian (5.2)–(5.3) satisfies every hypothesis of Theorem 6.2 on an ambient tensor product after adjoining one decoupled endpoint link. With*

$$\Delta = \min(1, h_E), \quad \tau = \frac{tm}{\Delta}, \tag{5.11}$$

therefore

$$|t| < t_{\text{FP}}^{(m)}(h_E) := \frac{\Delta}{m} \tau_{\text{FP}} \implies \text{gap}(H_{0,\text{dyn}} + tV)|_{\mathcal{H}_{\text{phys}}} \geq \frac{\Delta}{2}, \tag{5.12}$$

and the physical Gauss sector has a unique ground state. The bound is uniform in the number of clusters.

Proof. The audit is hypothesis by hypothesis: The appended endpoint link is decoupled and has a unique zero-flux ground state. The enlarged Hamiltonian is therefore the original ambient Hamiltonian tensored with this one-link factor; a lower bound on the enlarged gap is also a lower bound on the original ambient gap.

The normalized Hamiltonian is exactly (5.9) with $\bar{k} = 1$, so Theorem 6.2 gives a unique ambient ground state and normalized gap at least $1/2$. Every Gauss generator commutes with

FP hypothesis	DET chain realization	Result
Finite tensor-product cells	$\mathcal{H}_x = \mathcal{F}(\mathbb{C}^m) \otimes \mathbb{C}[\mathbb{Z}_{N_g}]$, of dimension $2^m N_g$ independent of chain length.	Pass
Identical one-site H_x	Put the outgoing link in cell x and append one decoupled final link, so every cell carries $q_{\phi,x} + h_E E_x^2$.	Pass
Rank-one local vacuum	$P_{\phi,x}$ and the zero-electric-flux projector both have rank one; their tensor product is the unique cell vacuum.	Pass
Normalized local gap	The cell gap is $\Delta = \min(1, h_E) > 0$; dividing H_x by Δ gives (5.8).	Pass
Fixed finite interval range	$K_{x,x+1}$ acts on two adjacent cells, hence $\bar{k} = 1$ independently of volume.	Pass
Ordinary tensor locality	In an ordered Jordan–Wigner representation the adjacent strings cancel; the even bilinear hopping, including its left-cell link operator, acts on exactly two cells.	Pass
Bounded self-adjoint interaction	$K_{x,x+1} = K_{x,x+1}^\dagger$ and $\ K_{x,x+1}\ = m$; thus $V_{x,x+1} = K_{x,x+1}/m$ has norm one.	Pass
Real small coupling	After factoring out Δ , the FP coupling is precisely $\tau = tm/\Delta$.	Pass
Gauss restriction	This is not an FP premise: all terms commute with every Gauss generator, and the ambient vacuum is physical because $m = 0 \pmod{N_g}$ with zero link flux.	Pass

Table 1: Fröhlich–Pizzo hypotheses audited against the finite-link DET chain.

the Hamiltonian. At $t = 0$ the unique ambient vacuum is in the trivial Gauss sector, since both $m = 4$ and $m = 16$ vanish modulo 2 and 4. Uniqueness and the discrete gauge eigenvalues keep the continued ground state in that sector. The physical spectrum is a subset of the ambient spectrum and contains its ground state, so restriction cannot reduce the excitation gap. Restoring the factor Δ proves (5.12). $\square$

6.1 What is and is not explicit in the FP window

The Lie–Schwinger proof contains one explicit series constant:

$$\frac{e^{8a} - 8a - 1}{a} + e^{8a} - 1 = 1, \quad a = 0.023320199830777650\dots \quad (5.14)$$

Appendix A of [4] uses the explicit convergence restriction

$$|\tau| < \frac{a}{16} = 0.001457512489423603\dots$$

Its local-gap estimate also has the explicit sufficient inequality

$$8|\tau| + 16|\tau| \sum_{\ell=3}^{\infty} \frac{|\tau|^{(\ell-2)/3}}{\ell} \leq \frac{1}{2}, \quad (5.15)$$

whose positive boundary is $\tau_{\text{gap}} = 0.046907169193748369\dots$. However, the induction closing Theorem 3.4 of [4] introduces unevaluated universal constants C, C' and repeatedly requires “sufficiently small” $|\tau|$. A scrupulous parameterization is therefore

$$\tau_{\text{FP}} = \min \left\{ \frac{a}{16}, \tau_{\text{gap}}, \tau_{\text{ind}} \right\} > 0, \quad (5.16)$$

where $\tau_{\text{ind}} > 0$ denotes those volume-independent but numerically unevaluated induction restrictions. Thus the two concrete model families have the cited, parametric windows

	$m = 4$ analogue	$m = 16$ DET16
$h_E = \frac{1}{2}$	$ t < \tau_{\text{FP}}/8$	$ t < \tau_{\text{FP}}/32$
$h_E \geq 1$	$ t < \tau_{\text{FP}}/4$	$ t < \tau_{\text{FP}}/16$

(5.17)

No numerical value of $t_{\text{FP}}^{(m)}$ follows unless τ_{ind} is evaluated.

The deterministic companion

`experiments/tfpt-discovery/ det16_dynamical_links_probe.py`

passes all 149 checks. It verifies exact Gauss invariance, commuting unperturbed terms, rank-one local vacua, the norms (5.5), the FP range and normalization hypotheses, the two explicit constants (5.14)–(5.15), the second-order effective Hamiltonian, and fourth-order finite-volume residual scaling. A long-range mutant supported on the two endpoints has diameter $L - 1$ and visibly fails the fixed- $\bar{k}$ premise. The smallest scanned gap remains 0.728854416 through $|t| = 0.4$ on two and three four-mode clusters. At $h_E \geq 1$, that scan point has normalized coupling $\tau = 1.6$ for $m = 4$ and $\tau = 6.4$ for $m = 16$, both far outside the cited window; the scan is evidence against a finite-volume closing there, not an extension of (5.12). The finite $\mathbb{Z}_2/\mathbb{Z}_4$ chain item is therefore cited-verified with the parametric FP window, not with an invented numerical threshold.

7 The proposed explicit count and why it is conditional

Proposition 7.1 (Elementary relative-form gap bound). *Let $P = |\Omega_{\Lambda, \phi}\rangle\langle\Omega_{\Lambda, \phi}|$ and $Q = \mathbf{1} - P$. Suppose a self-adjoint perturbation V satisfies*

$$PVQ = QVP = 0, \quad PVP = v_0 P, \quad (6.1)$$

and

$$|\langle\psi, (V - v_0 \mathbf{1})\psi\rangle| \leq c \langle\psi, H_0 \psi\rangle \quad \text{for every } \psi \in Q\mathcal{H}. \quad (6.2)$$

Then the ground vector remains $|\Omega_{\Lambda, \phi}\rangle$, and

$$\text{gap}(H_0 + tV) \geq 1 - c|t| \geq 1 - 2c|t|. \quad (6.3)$$

In particular, the weaker displayed bound is positive for $|t| < t_{0, \text{rel}} := 1/(2c)$.

Proof. Condition (6.1) makes $P\mathcal{H}$ and $Q\mathcal{H}$ reducing subspaces. The energy on $P\mathcal{H}$ is tv_0 . For normalized $\psi \in Q\mathcal{H}$, Theorem 2.2 gives $\langle\psi, H_0 \psi\rangle \geq 1$, and

$$\begin{aligned} \langle\psi, (H_0 + tV - tv_0 \mathbf{1})\psi\rangle &\geq (1 - c|t|) \langle\psi, H_0 \psi\rangle \\ &\geq 1 - c|t|. \end{aligned}$$

The min–max principle proves the first inequality in (6.3); the second is a weakening for $c|t| \geq 0$. $\square$

7.1 The norm count

For unit-amplitude nearest-neighbour hopping across one bond,

$$K_{xy} = \sum_{\alpha, \beta=1}^{16} \left(a_{x,\alpha}^\dagger U_{\alpha\beta} a_{y,\beta} + a_{y,\beta}^\dagger U_{\alpha\beta}^* a_{x,\alpha} \right), \quad (6.4)$$

with U unitary, the one-particle bond matrix has sixteen eigenvalues $+1$ and sixteen eigenvalues -1 . Its fermionic second quantization therefore has

$$\|K_{xy}\| = 16. \quad (6.5)$$

If the lattice coordination is z , at most z such bonds touch one cluster. The advertised count is consequently

$$c_{\text{count}} = 16z, \quad t_{0,\text{count}} = \frac{1}{32z}. \quad (6.6)$$

On a three-dimensional cubic spatial slice, $z = 6$, so

$$c_{\text{count}} = 96, \quad t_{0,\text{count}} = \frac{1}{192} = 0.005208333\dots \quad (6.7)$$

This count controls the norm incident on one site. It does not prove (6.1) or (6.2). In fact ordinary hopping creates a pair of defective clusters from components of the product ground state, so

$$QKP \neq 0. \quad (6.8)$$

The numeric twin measures $\|QKP\| = \sqrt{2}$ on the two-cluster four-mode analogue. Thus Proposition 7.1 cannot be applied to (6.4), and (6.7) is not the explicit t_* of Theorem 5.1. A global Weyl estimate would only give $1 - 2|t|\|V\|$, whose right side scales with volume and is unsuitable.

Uncertified explicit-threshold box. The number $t_{0,\text{count}} = 1/192$ is the value obtained from the requested local norm count on a cubic spatial slice. It is a valid threshold for the *conditional* relative-form proposition if (6.1)–(6.2) are separately proved with $c = 96$. Ordinary hopping fails (6.1), visibly and numerically. The applicable Michalakis–Zwolak theorem instead gives $t_* = J_0/J_V > 0$ with $J_0 = 1/(3c_{\text{MZ}})$, where c_{MZ} contains the quasi-local spectral-flow decay w in (4.9). No value of c_{MZ} has been derived for this model, so replacing it by 96 would be an unjustified theorem claim.

8 Numeric twin

The companion probe is

`experiments/tfpt-discovery/det16_gap_stability_probe.py`.

It uses the exact four-mode determinant projectors and open-chain hopping machinery of the frozen v1002 route. All 19 checks pass. The direct comparison is:

clusters	$t = 0.05$	$t = 0.10$	$t = 0.20$	$t = 0.30$
2	0.973726995	0.946980790	0.902548402	0.837138848
3	0.962836264	0.924929608	0.860366685	0.764340648
4	0.957577627	0.914546582	0.842484121	0.726091276

(7.1)

Every entry lies above $1 - 2c_{\text{count}}|t|$ with $c_{\text{count}} = 4$ for the two-cluster endpoint pair and 8 for the three–four cluster chains. Except at the smallest t , this linear line is already negative and hence not tight.

The scan uses step 0.05 for two and three clusters and step 0.10 for four clusters on $0 \leq t \leq 4$. It finds no finite-volume gap closing. The minimum scanned gaps are

clusters	min gap	minimizing t
2	0.764490286	0.55
3	0.472500856	4.00
4	0.549772255	0.60

(7.2)

Consequently the requested finite-chain “actual closing” t_c is not located because it does not occur in the declared scan:

$$t_c^{\text{scan}} > 4, \quad \frac{t_{0,\text{count}}^{\text{toy}}}{t_c^{\text{scan}}} < \frac{1/16}{4} = 0.015625. \quad (7.3)$$

This is an upper bound on a ratio, not an invented critical point.

The positive and negative controls expose the theorem hypotheses:

$$\begin{aligned}
\text{Spin}(4)\text{-covariant flavor-diagonal hopping:} & \max_A \left\| [K, G^A] \right\| = 0, \\
\text{flavor-selective mutant:} & \max_A \left\| [K_{\text{mut}}, G^A] \right\| = 1, \\
1/3 \text{ normalization mutant:} & \left\| \tilde{P}^2 - \tilde{P} \right\| = 1/9, \quad \text{Tr } \tilde{P} = 2/3.
\end{aligned} \quad (7.4)$$

Thus one mutant visibly leaves the covariant perturbation class and the other destroys the projector/frustration-free premise.

9 Combined strongest theorem

Theorem 9.1 (Volume-independent DET16 mirror-gap theorem). *For the matter-only DET16 cluster model (1.1)–(1.6) on any finite bounded-degree lattice:*

- (a) *the projectors are mutually commuting even CAR operators; H_0 has exact spectrum $\{0, \dots, |\Lambda|\}$, multiplicities (2.2), unique product ground state (2.3), and gap one in every volume;*
- (b) *H_0 is locally Spin(10)-Gauss invariant, parity invariant, and translation invariant for constant ϕ ; its unique product ground state has neither spontaneous symmetry breaking nor torus/topological degeneracy;*
- (c) *every uniformly bounded finite-range even perturbation has a volume-independent stability interval $|t| \leq t_*$ on which the gap is at least $1/2$, with $t_* > 0$ supplied by the Michalakis–Zwolak theorem after the exact checks $\Delta_0 = 0$ and $\gamma(r) = 1$;*
- (d) *for finite $\mathbb{Z}_2/\mathbb{Z}_4$ links on an open chain, $h_E > 0$, and gauge-covariant nearest-neighbour hopping, the physical Gauss sector has a unique ground state and volume-independent gap at least $\Delta/2$ throughout the cited window $|t| < \Delta\tau_{\text{FP}}/16$, with $\Delta = \min(1, h_E)$ and $\tau_{\text{FP}} > 0$ parameterized by (5.16);*
- (e) *if, additionally, a perturbation obeys the block-diagonal relative-form hypotheses (6.1)–(6.2), then the explicit elementary estimate (6.3) holds.*

No numerical identification of t_ with $t_{0,\text{count}} = 1/(32z)$ is asserted.*

Proof. Part (a) is Theorem 2.2 and Lemma 2.1. Part (b) is Theorem 3.2. Part (c) is Theorem 5.1, whose proof verifies every hypothesis of the quoted stability theorem against the concrete product model. Part (d) is Proposition 6.3, including the ambient-to-physical gap restriction. Part (e) is Proposition 7.1. The final sentence follows from (6.8) and the explicit-threshold box. $\square$

10 Physical spectator and remaining boundary

10.1 Exact spectator statement

Let $\mathcal{H}_{\text{wall}}$ carry a wall Hamiltonian H_{wall} and let $\mathcal{H}_{\text{mir}}$ carry $H_{0,\Lambda}$. If the DET16 terms act as the identity on wall modes, then

$$H_{\text{total}}(0) = H_{\text{wall}} \otimes \mathbf{1}_{\text{mir}} + \mathbf{1}_{\text{wall}} \otimes H_{0,\Lambda}. \quad (9.1)$$

Hence

$$\text{spec } H_{\text{total}}(0) = \{E_{\text{wall}} + k : E_{\text{wall}} \in \text{spec } H_{\text{wall}}, k = 0, \dots, |\Lambda|\}. \quad (9.2)$$

The $k = 0$ copy is exactly the original wall spectrum, with every multiplicity preserved.

For $t \neq 0$, “does not mix” must mean that no interaction term contains both wall and mirror operators. Exact preservation, rather than merely perturbative continuity, further requires

$$V = \mathbf{1}_{\text{wall}} \otimes V_{\text{mir}}. \quad (9.3)$$

Then the total Hamiltonian remains a tensor sum, and the stable mirror ground band adds one common scalar energy to the wall spectrum. If a wall-only term is allowed it changes the wall spectrum on its own; if a wall–mirror coupling is allowed, (9.2) is no longer exact and a separate relative/local stability estimate is required.

Remaining boundary for CHIRAL4D.MIRROR.DET16.01.

Compact/continuous dynamical gauge fields. Section 6 closes the finite $\mathbb{Z}_2/\mathbb{Z}_4$ open-chain item by Theorem 6.2 and Proposition 6.3; it is no longer part of this boundary box. Compact-group rotor links have infinite-dimensional local Hilbert spaces and therefore violate the finite- M hypothesis of the bounded-interaction FP theorem quoted here. They require the form-bounded extension with its domain hypotheses, or truncation-uniform estimates, neither of which is audited in this memorandum.

Domain wall. At $t = 0$ the spectator theorem (9.1)–(9.2) is exact. For $|t| < t_*$, exact preservation of the wall spectrum requires the factorization (9.3). Gauge covariance or fermion parity alone does not forbid wall–mirror bilinears, so the microscopic domain-wall construction must prove their absence or bound them.

Three spatial dimensions. The count $z = 6$ and $1/192$ assumes a cubic three-dimensional spatial slice, but no lattice domain-wall Hamiltonian, chirality localization, continuum limit, or Lorentz-covariant $3 + 1$ -dimensional theory is constructed here. The theorem is a uniform quantum-lattice statement for the mirror matter sector, not yet the physical four-dimensional placement.

Explicit hopping threshold. The applicable stability theorem proves $t_* > 0$ and gap at least $1/2$, but its constant contains the spectral-flow decay c_{MZ} . The simple value $1/192$ is conditional and cannot be promoted to the hopping theorem because $QKP \neq 0$. Deriving a numerical c_{MZ} , or proving a different explicit local block-diagonalization/Knabe estimate, remains open.

Conclusion

T1 and T2 are elementary, exact, and volume independent. T3 is a genuine thermodynamic stability statement: the product projector has zero Local-TQO error and unit local gap, so the full Michalakis–Zwolak theorem applies to bounded finite-range even perturbations. Its threshold is uniform but not numerically evaluated. The requested count $1/(32z)$ is retained only in the precise conditional proposition it actually satisfies; the numeric twin both passes the comparison and detects the missing block-diagonal premise. The dynamical-link audit repairs the tensor-product endpoint exactly and applies the Fröhlich–Pizzo theorem, which does control the spectral gap uniformly in chain length. This moves finite $\mathbb{Z}_2/\mathbb{Z}_4$ open chains to cited-verified status with the parametric window $|t| < \Delta\tau_{\text{FP}}/m$; it does not manufacture a numerical value for the paper’s implicit induction constant. Accordingly, the red box now retains only compact/continuous links, domain-wall mixing, and the $3 + 1$ -dimensional placement, preventing the chain theorem from being mistaken for the complete dynamical-gauge contract.

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